THE BLOW-UP SQUARE YIELDS A LONG EXACT COHOMOLOGY SEQUENCE FOR TORSION COEFFICIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the descent triangle and the identification of the cohomology of the three terms through the section formula for a pushforward.

Editorial note. The label lemma-blow-up-square-etale-cohomology names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's blow-up square is the author's and is delivered verbatim; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The cohomological descent triangle of the blow-up square. Same chapter, the lemma producing the distinguished triangle K -> Ri_*(K|_Z) + Rb_*(K|_{X'}) -> Rc_*(K|_E) -> K[1]. Supplies the triangle whose long exact cohomology sequence is the assertion.
- Derived global sections of a pushforward. Cohomology on Sites, the Leray section describing R Gamma(X, Rb_* of a complex) as R Gamma(X' , -). Supplies the identification of the cohomology groups of the three terms, and similarly for the other pushforwards.

Source reference. Original reference 0EW5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 23315–23315. Prerequisite pointer invoked at proof line(s) 23411; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 072X: sources/stacks-chow/tags.txt, source lines 9041–9041. Prerequisite pointer invoked at proof line(s) 23416; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme and let $K \in D^+(X_{étale})$ have torsion cohomology sheaves. Let $Z \subset X$ be a closed subscheme cut out by a quasi-coherent ideal of finite*

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type. Consider the corresponding blow up square

$$\begin{array}{ccc} E & \longrightarrow & X' \\ \downarrow & & \downarrow b \\ Z & \longrightarrow & X \end{array}$$

Then there is a canonical long exact sequence

$$H^p_{étale}(X, K) \to H^p_{étale}(X', K|_{X'}) \oplus H^p_{étale}(Z, K|_Z) \to H^p_{étale}(E, K|_E) \to H^{p+1}_{étale}(X, K)$$

First proof. This follows immediately from Lemma 0EW5 and the fact that

$$R\Gamma(X, Rb_*(K|_{X'})) = R\Gamma(X', K|_{X'})$$

(see Cohomology on Sites, Section 072X) and similarly for the others. □